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Remote Control of the Motion of a Differential Drive Robot by Moving the Palm of the Hand

Semester Project Report
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Abstract

This project presents the design and implementation of a hand-gesture-controlled differential drive mobile robot. The system allows the robot to be controlled remotely by moving the palm of the hand, providing a simple and intuitive alternative to traditional control methods such as wired controllers or joysticks. The project is based on two main units: a transmitter unit mounted on the hand and a receiver unit installed on the robot.

The transmitter unit consists of an Arduino Nano, an MPU-6050 accelerometer and gyroscope sensor, an HC-05 Bluetooth module, and a portable power supply. The MPU-6050 sensor detects the tilt and movement of the user's palm along different axes. These readings are processed by the Arduino Nano and converted into movement commands such as forward, backward, right, left, and stop. The commands are then transmitted wirelessly through the HC-05 Bluetooth module.

The receiver unit consists of another Arduino Nano, an HC-05 Bluetooth module, an L298 motor driver, DC motors, and a battery supply. After receiving the command from the transmitter, the Arduino Nano controls the L298 motor driver to operate the motors according to the required direction. As a result, the robot moves forward, backward, turns right, turns left, or stops depending on the hand gesture.

The system was programmed using the Arduino IDE. The final implementation showed that the robot successfully responded to palm movements, moved in the main directions, and stopped when the hand returned to its neutral position. This project demonstrates the integration of sensors, wireless communication, embedded programming, and motor control in one practical robotic system. It also provides a useful foundation for future applications in assistive robotics, remote control systems, and gesture-based human–machine interaction.

Keywords

Hand Gesture Control, Differential Drive Robot, Arduino, MPU-6050, Bluetooth Communication, L298 Motor Driver, Mobile Robotics, Embedded Systems

الملخص

يقدّم هذا المشروع تصميم وتنفيذ روبوت متحرك تفاضلي يتم التحكم بحركته عن بُعد من خلال حركة راحة اليد. يهدف النظام إلى توفير طريقة تحكم بسيطة وتفاعلية بديلة عن وسائل التحكم التقليدية مثل أجهزة التحكم السلكية أو عصا التحكم. يعتمد المشروع على وحدتين أساسيتين: وحدة إرسال يتم تثبيتها على اليد، ووحدة استقبال يتم تركيبها على الروبوت.

تتكون وحدة الإرسال منArduino Nano ، وحساس الحركة MPU-6050 الذي يحتوي على مقياس تسارع وجيروسكوب، بالإضافة إلى وحدة Bluetooth من نوع HC-05 ومصدر تغذية محمول. يقوم حساس MPU-6050 بقراءة ميلان وحركة راحة اليد على المحاور المختلفة، ثم يقوم Arduino Nanoبمعالجة هذه القراءات وتحويلها إلى أوامر حركة مثل التقدم للأمام، الرجوع للخلف، الالتفاف إلى اليمين، الالتفاف إلى اليسار، أو التوقف. بعد ذلك، يتم إرسال هذه الأوامر لاسلكياً إلى الروبوت عبر وحدةBluetooth.

أما وحدة الاستقبال فتتكون من Arduino Nano آخر، ووحدةHC-05 Bluetooth ، ودرايفر محركاتL298 ، ومحركات تيار مستمر، بالإضافة إلى مصدر تغذية مناسب. بعد استقبال الأمر المرسل من وحدة اليد، يقوم Arduino Nano في الروبوت بتشغيل درايفر المحركات L298 للتحكم باتجاه دوران المحركات حسب الأمر المطلوب، مما يسمح للروبوت بالحركة للأمام أو الخلف أو الالتفاف يميناً ويساراً أو التوقف.

تمت برمجة النظام باستخدام بيئة .Arduino IDE أظهرت النتائج أن الروبوت استجاب بنجاح لحركات راحة اليد، وتمكن من الحركة في الاتجاهات الأساسية، كما توقف عند عودة اليد إلى الوضع الطبيعي. يوضح هذا المشروع إمكانية دمج الحساسات، والاتصال اللاسلكي، والبرمجة المدمجة، والتحكم بالمحركات ضمن نظام روبوتي عملي واحد. كما يشكل أساساً مناسباً لتطبيقات مستقبلية في الروبوتات المساعدة، وأنظمة التحكم عن بُعد، والتفاعل بين الإنسان والآلة باستخدام الإيماءات.

الكلمات المفتاحية

التحكم بالإيماءات اليدوية، روبوت الدفع التفاضلي، أردوينو، حساسMPU-6050 ، الاتصال عبر البلوتوث، درايفر المحركاتL298 ، الروبوتات المتنقلة، الأنظمة المدمجة

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Mathematical Symbols

Symbol	Meaning	Unit / Description
v	Linear velocity of the robot	m/s
ω	Angular velocity of the robot	rad/s
VR	Right wheel linear velocity	m/s
VL	Left wheel linear velocity	m/s
L	Distance between the two driving wheels	m
X	Robot position on the x-axis	m
Y	Robot position on the y-axis	m
θ	Robot orientation angle	rad or degree
dot{x}	Rate of change of robot position on x-axis	m/s
dot{y}	Rate of change of robot position on y-axis	m/s
dot{ θ}	Rate of change of robot orientation	rad/s
ax	MPU-6050 accelerometer reading on x-axis	Raw value / LSB
ay	MPU-6050 accelerometer reading on y-axis	Raw value / LSB
θx	Hand tilt angle around x-axis	degree
θy	Hand tilt angle around y-axis	degree

-16384	MPU-6050 accelerometer sensitivity at ±2g	LSB/g
π	Mathematical constant pi	≈ 3.1416
Kx	Proportional gain for turning control	Used in improved model
Ky	Proportional gain for forward/backward speed control	Used in improved model
F	Forward command	Robot moves forward
B	Backward command	Robot moves backward
R	Right command	Robot turns right
L_c	Left command	Robot turns left
S	Stop command	Robot stops

List of Abbreviations

Abbreviation	Full Form
ADXL335	Analog Devices 3-Axis Accelerometer ADXL335
AI	Artificial Intelligence
Arduino IDE	Arduino Integrated Development Environment
B	Backward Command
C++	C Plus Plus Programming Language
CNN	Convolutional Neural Network
DC	Direct Current
F	Forward Command
HC-05	HC-05 Bluetooth Serial Communication Module
I2C	Inter-Integrated Circuit
IMU	Inertial Measurement Unit
IoT	Internet of Things
IR	Infrared
KPI	Key Performance Indicator
L	Left Command

L298	L298 Dual H-Bridge Motor Driver
LSB	Least Significant Bit
MQTT	Message Queuing Telemetry Transport
MPU-6050	Motion Processing Unit 6050
PWM	Pulse Width Modulation
R	Right Command
RF	Radio Frequency
RX	Receiver / Receive Pin
S	Stop Command
TX	Transmitter / Transmit Pin
USB	Universal Serial Bus
Wi-Fi	Wireless Fidelity

Chapter 1
General Introduction

1.1 Project Overview and Background

This project focuses on the design and implementation of a differential drive mobile robot equipped with two driving wheels and a free front wheel, controlled entirely through hand gestures using a wearable glove. The glove-mounted unit consists of an Arduino Nano microcontroller and an MPU-6050 accelerometer and gyroscope sensor, which together capture the tilt and orientation of the user's palm. The hand gestures are mapped to specific movement commands: tilting the hand upward commands the robot to move forward, tilting downward commands backward motion, tilting right commands a right turn, tilting left commands a left turn, and returning the hand to the neutral position stops the robot.

The system is divided into two main units: a transmitter unit on the hand and a receiver unit on the robot. The transmitter processes the sensor data and sends the corresponding commands wirelessly via an HC-05 Bluetooth module. The receiver unit, consisting of another Arduino Nano, an HC-05 Bluetooth module, an L298 motor driver, and DC motors powered by a battery, interprets the commands and controls the differential drive motors accordingly. This setup ensures intuitive, real-time control of the mobile robot based solely on natural hand gestures. The design integrates sensors, wireless communication, embedded programming, and motor control into a single practical system, providing a foundation for future research in assistive robotics and gesture-based human–machine interaction [1][2].

1.2 Problem Statement and Significance

Elderly individuals and people with disabilities often face difficulties when using conventional control systems for mobile robots or automated devices. Existing human–robot interaction systems that provide assistance in daily tasks are limited, and their interfaces are typically not intuitive or accessible. Moreover, the use of

robots for supportive daily activities remains scarce due to the complexity of control methods and the physical effort required. There is a clear need for simple, natural, and low-effort control mechanisms to improve the quality of life for these user groups.

This project addresses these challenges by implementing a hand-gesture-controlled mobile robot that integrates wearable sensors, wireless communication, and embedded processing. By allowing intuitive control through natural hand movements, the system simplifies interaction, reduces the learning curve, and enhances accessibility. The significance of this work extends to demonstrating practical applications of sensor technologies, reinforcing the concept of human–machine interaction, and providing an educational platform for engineering design and development. Furthermore, the system establishes a foundation for future enhancements and potential applications in assistive robotics, remote control, and gesture-based human–robot collaboration [1][2].

1.3 Project Objectives

1.3.1 Main Objective

The primary goal of this project is to design and implement a hand-gesture-controlled mobile robot that enables intuitive and natural interaction between humans and the robot, providing an accessible solution for elderly users and individuals with disabilities [1][2].

1.3.2 Specific Objectives

1. **Gesture-Based Control:** Develop a system that accurately interprets hand movements and translates them into robot motion commands for forward, backward, right, left, and stop operations.

2. **Real-Time Response:** Ensure the robot responds promptly and precisely to hand gestures, minimizing latency in motion control [2].
3. **Reliable Wireless Communication:** Implement a robust wireless link between the transmitter (hand unit) and the robot, using HC-05 Bluetooth modules for low-latency, dependable communication [1].
4. **Ease of Use and Accessibility:** Design the system interface to be simple and intuitive, allowing elderly and disabled users to operate the robot effectively without extensive training.
5. **Future Expandability:** Create a scalable platform that allows integration of additional modules, such as a robotic arm for assistive tasks, without major redesign.

These objectives are structured according to the **SMART** principles (Specific, Measurable, Achievable, Relevant, Time-bound), ensuring that each target is practical, assessable, and contributes directly to the overall goal of improving human–robot interaction [1][2].

1.4 Definitions and Key Concepts

1. **Gesture:** A gesture is a deliberate movement of the hand or body used to convey information or commands. In the context of robotics, hand gestures are used to control the motion and actions of mobile robots through wearable sensors [1][2].

2. **Differential Drive Robot:** A differential drive robot is a mobile robot that uses two independently driven wheels to achieve movement and steering. Motion is controlled by varying the speed and direction of each wheel, allowing forward, backward, and turning maneuvers [1].

3. **MPU-6050 Sensor:** The MPU-6050 is a 6-axis inertial measurement unit (IMU) that combines a 3-axis accelerometer and a 3-axis gyroscope. It measures the tilt, acceleration, and orientation of the hand and sends the data to a microcontroller for processing [2].
4. **Arduino Nano:** Arduino Nano is a compact microcontroller board based on the ATmega328, used for embedded programming and sensor data processing. It receives inputs from sensors like MPU-6050, executes control algorithms, and sends commands to actuators such as motor drivers [1][2].
5. **HC-05 Bluetooth Module:** The HC-05 is a Bluetooth serial communication module used for wireless data transmission between the transmitter unit (hand) and the receiver unit (robot). It provides low-latency and reliable communication for real-time robot control [1].

6. **L298 Motor Driver:** L298 is a dual H-bridge motor driver integrated circuit that allows the Arduino to control the direction and speed of two DC motors simultaneously. It receives logic commands and amplifies current to drive the motors [1].

7. **Embedded System:** An embedded system is a dedicated computing system designed to perform specific control functions within a larger system. In this project, it refers to the combination of Arduino, sensors, and motor drivers that enable gesture-based robot control [2].

1.5 Project Scope and Technical Requirements

1.5.1 Project Scope

The scope of this project includes the design, development, and implementation of a two-wheeled differential drive mobile robot with a free front wheel, controlled entirely via hand gestures. The system encompasses a wearable hand unit (transmitter) and a robot-mounted receiver unit. The transmitter captures palm movements through an MPU-6050 sensor and Arduino Nano, and communicates commands wirelessly using HC-05 Bluetooth. The receiver unit interprets these commands through another Arduino Nano, controlling the L298 motor driver and DC motors to execute forward, backward, left, right, and stop movements. The system focuses on intuitive and accessible control for users, particularly the elderly and individuals with disabilities, while providing a foundation for future expansion, such as integrating a robotic arm for assistive tasks [1][2].

1.5.2 Technical Requirements

Table 1.1: Project Technical Specifications

Category	Parameter	Specification
Weight	Robot Weight	≤ 200 g (Target: 150–200 g)
Weight	Controller/Transmitter Weight	≤ 120 g (Target: 80–120 g)
Speed	Maximum Speed	0.25 – 0.35 m/s (≈ 0.9 – 1.2 km/h)
Operating Range	Wireless Range	≥ 10 m indoor / up to 20–30 m outdoor
Drive System	Robot Mobility	Differential drive (2 wheels + free caster)

Motion Functions	Movement Modes	Forward – Backward – Left – Right – Stop
Sensing System	Motion Detection	Accurate hand motion detection via MPU-6050
Power System	Power Source	Rechargeable lithium battery for both units
Performance	Response Time	< 1 second for gesture recognition and motor activation
Performance	Operating Time	20–30 minutes per full charge
Communication	Wireless Link	Stable HC-05 Bluetooth connection

1.5.3 Additional Considerations

- Safety mechanisms to prevent erratic motion when communication is lost [2].
- Compact and lightweight design to facilitate wearability and mobility [1].
- Modularity to allow future upgrades, such as adding a robotic arm for assistive tasks or sensors for obstacle detection [2].

1.6 Project Contribution and Novelty

This project contributes to the field of human–robot interaction by implementing a practical, gesture-based control system for a differential drive mobile robot. Unlike previous studies [1][2] that focused on either accelerometer-based or vision-based gesture recognition, this system integrates a wearable MPU-6050 sensor, Arduino Nano microcontrollers, HC-05 Bluetooth communication, and L298 motor drivers into a compact and reliable solution. The robot responds intuitively and in real-time to natural hand movements, providing an accessible interface for elderly users and individuals with disabilities.

The novelty of this work lies in the combination of accuracy, low-latency response, and user-centered design. The system demonstrates how wearable sensor technology can be applied effectively in mobile robotics to achieve intuitive control without complex hardware or extensive training. Additionally, the platform is designed to be scalable, allowing future enhancements such as the integration of a robotic arm for assistive tasks or additional sensors for obstacle detection. This contribution extends the practical application of gesture-based robotics beyond laboratory settings, offering a foundation for future research in assistive robotics, education, and remote control systems [1][2].

1.7 Design Methodology

The design methodology of this project follows a systematic approach to develop a robust gesture-controlled mobile robot. Initially, a differential drive platform was selected for its simplicity, maneuverability, and ease of integration with sensors and motor drivers [1]. The hand-mounted transmitter unit was designed to be lightweight and wearable, using an Arduino Nano and MPU-6050 sensor to capture hand orientation and motion accurately.

Sensor data is processed in real-time by the Arduino, which translates palm movements into five primary robot commands: forward, backward, left, right, and stop. Wireless communication is achieved via HC-05 Bluetooth modules to ensure low-latency, reliable data transmission between the transmitter and receiver [1][2]. The receiver unit includes an Arduino Nano and L298 motor driver to actuate the robot's differential drive motors based on the received commands.

A modular and iterative approach was adopted during the design and testing phases, allowing continuous refinement of sensor calibration, wireless reliability, and motor control algorithms. Each subsystem was tested independently before integration to ensure accuracy, responsiveness, and robustness. This methodology ensures that the

final system provides intuitive control, practical usability, and scalability for future enhancements such as adding a robotic arm or additional sensors [1][2].

1.8 Report Structure

This report is structured to provide a comprehensive overview of the hand-gesture-controlled differential drive robot project, from theoretical background to implementation and testing.

- **Chapter 1 – General Introduction:** Introduces the project, its objectives, problem statement, scope, and contributions, along with key definitions and design methodology.
- **Chapter 2 – Literature Review and Previous Work:** Presents a detailed review of existing research on gesture-controlled robotics, sensor-based and vision-based recognition techniques, and wireless communication methods, highlighting the gap addressed by this project [1][2].
- **Chapter 3 – Project Analysis and Design:** Describes the system architecture, block diagrams, hardware design, sensor integration, motor control logic, and wireless communication framework.
- **Chapter 4 – Implementation and Testing:** Provides details on hardware assembly, software programming, calibration, and testing results of gesture recognition and robot motion.
- **Chapter 5 – Conclusion and Recommendations:** Summarizes the project achievements, discusses limitations, and presents suggestions for future work and system enhancements.
- **Appendices:** Include full technical specifications, schematics, block diagrams, code listings, and other supplementary materials supporting the report.

This structured approach ensures clarity, logical flow, and easy understanding for the reader, providing a comprehensive presentation of the project's design, development, and evaluation process [1][2].

1.9 Chapter Conclusion

This chapter provided an overview of the hand-gesture-controlled differential drive robot project, highlighting the project's objectives, problem statement, scope, technical requirements, contributions, and design methodology. Key definitions and concepts were introduced to establish a clear understanding of the system components and their roles. The report structure was also outlined to guide the reader through the subsequent chapters. The chapter demonstrates the significance of developing an intuitive and accessible robot control system, laying the foundation for the detailed literature review and design analysis presented in Chapters 2 and 3 [1][2].

Chapter 2

Review of literature and previous studies

2.1 Previous Studies

Several recent studies have explored gesture-based control of mobile robots, providing valuable insights for the development of intuitive human–robot interaction systems. Vala (2018) proposed a wireless robot controlled by hand gestures using an ADXL335 accelerometer and Arduino Uno. The system was divided into a transmitter unit on the hand and a receiver unit on the robot, with five recognized gestures: forward, backward, left, right, and stop [1]. This study demonstrated the feasibility of accelerometer-based gesture control but relied on RF communication and older sensor technology [1].

Rohan et al. (2024) integrated wearable sensors and IoT frameworks to achieve real-time gesture control. Their system combined sensor-based recognition (MPU-6050) with optional vision-based methods (camera + CNN), using MQTT protocols for command transmission to mobile robots [2]. This approach provided accurate gesture detection, scalability, and real-time responsiveness, though vision-based methods introduced higher computational demands [2].

These studies provide the theoretical and practical basis for this project. The transmitter–receiver concept, gesture mapping, and the use of accelerometer data are adopted, while our system improves on communication reliability, latency, and system simplicity by using MPU-6050 sensors, HC-05 Bluetooth, and L298 motor drivers for a differential drive robot [1][2].

2.2 Comparative Analysis of Existing Systems

Table 2.1: Comparative Analysis of Existing Gesture-Controlled Robots

System / Study	Technology / Sensors	Communication	Strengths	Limitations	Application to Project	Reference
Vala, 2018	ADXL335 accelerometer + Arduino Uno	RF	Lightweight, simple gesture mapping, 5 gestures recognized	Older sensor technology, RF prone to interference	Adopt transmitter–receiver design and gesture mapping, but update to MPU-6050 & Bluetooth for reliability	[1]
Rohan et al., 2024	MPU-6050 sensor + Camera + CNN	MQTT / IoT	Accurate detection, scalable, real-time, supports hybrid approaches	Vision-based methods require high computation, sensitive to lighting	Use sensor-based recognition only (MPU-6050) for low-cost, reliable real-time control	[2]
Other Vision-Based Systems	Camera, image processing algorithms	Wi-Fi / wired	High accuracy under controlled conditions, supports complex gestures	Sensitive to lighting and background, high processing requirements	Not adopted; focus on lightweight wearable sensor-based control	[2]

2.3 Research Gap and Proposed Innovation

Although previous studies have demonstrated the feasibility of gesture-based robot control using accelerometers and vision-based systems, several gaps remain. The system proposed by Vala (2018) relies on RF communication and older accelerometer technology, which limits reliability and scalability [1]. On the other hand, advanced IoT-based systems, such as that by Rohan et al. (2024), integrate sensor and vision-based recognition for real-time control, but they require complex computation and higher-cost hardware [2].

The innovation of this project lies in combining the benefits of both approaches while addressing their limitations. By using a wearable MPU-6050 sensor, HC-05 Bluetooth modules, and L298 motor drivers, the system achieves accurate, low-latency, and reliable gesture-based control without the need for heavy computational resources or high-cost vision systems. Additionally, the design emphasizes accessibility, making it suitable for elderly users and individuals with disabilities, which has not been fully addressed in previous studies. This approach provides a practical, scalable, and low-cost solution for differential drive mobile robots, laying the foundation for future enhancements such as integrating a robotic arm or additional sensors [1][2].

2.4 Theoretical Framework

The theoretical framework for this project is based on integrating mechanical, electrical, and computational components to enable accurate and intuitive gesture-based control of a differential drive mobile robot. Mechanically, the robot consists of a two-wheeled differential drive platform with a free caster wheel, ensuring stability and maneuverability [1].

Electrically, the system employs an Arduino Nano microcontroller on both the transmitter (hand) and receiver (robot) units, with an MPU-6050 sensor to capture hand orientation and motion [2]. The L298 motor driver is used to actuate the differential drive motors, allowing precise control of forward, backward, left, and right movements. Wireless communication is facilitated by HC-05 Bluetooth modules, providing reliable and low-latency transmission of gesture commands between the transmitter and receiver [1][2].

From a computational perspective, the Arduino IDE is used for embedded programming, allowing real-time processing of sensor data and execution of motor control logic. This framework integrates the principles of sensor-based gesture recognition, embedded systems, and motor control to achieve responsive and user-friendly robot operation. Additionally, the design supports scalability for future enhancements, such as integrating additional sensors or a robotic arm for assistive tasks [1][2].

2.5 Chapter Conclusion

This chapter presented a comprehensive review of existing literature and previous research related to gesture-controlled mobile robots. Key studies, including Vala (2018) and Rohan et al. (2024), were analyzed to identify their technologies, communication methods, strengths, and limitations [1][2]. A comparative analysis highlighted the advantages and constraints of accelerometer-based, vision-based, and hybrid systems, providing a clear rationale for the design decisions in the current project.

The theoretical framework was outlined, detailing the mechanical, electrical, and computational components essential for accurate, low-latency, and reliable gesture-based control. The chapter also identified the research gap, emphasizing the need for a low-cost, accessible, and user-friendly system suitable for elderly users and

individuals with disabilities. Overall, this review establishes the foundation for the design, implementation, and evaluation of the hand-gesture-controlled differential drive robot, guiding the methodology presented in the subsequent chapters [1][2].

Chapter 3

Project Analysis and Design

3.1 System Architecture

The system architecture of the proposed hand-gesture-controlled differential drive robot is based on two main subsystems: a transmitter unit and a receiver unit. The transmitter unit is designed as a wearable hand glove and consists mainly of an MPU-6050 sensor, an Arduino Nano, and an HC-05 Bluetooth module. Its main function is to detect the tilt and movement of the user's hand, process the sensor readings, and convert them into motion commands such as forward, backward, right, left, and stop. These commands are then transmitted wirelessly through Bluetooth to the receiver unit [1][2].

The receiver unit is mounted on a mobile robot car kit that uses a differential drive mechanism consisting of two driving wheels and one free caster wheel. It includes an Arduino Nano, an HC-05 Bluetooth module, an L298 motor driver, and gear motors. The receiver unit accepts the transmitted commands and controls the motors accordingly to execute the required robot motion. Communication between the MPU-6050 sensor and the transmitter Arduino is achieved through the I2C protocol, while wireless communication between the transmitter and receiver is established through the two HC-05 Bluetooth modules. The overall system is programmed using Arduino IDE in C++, providing a simple and effective embedded control structure for gesture-based robot operation [1][2].

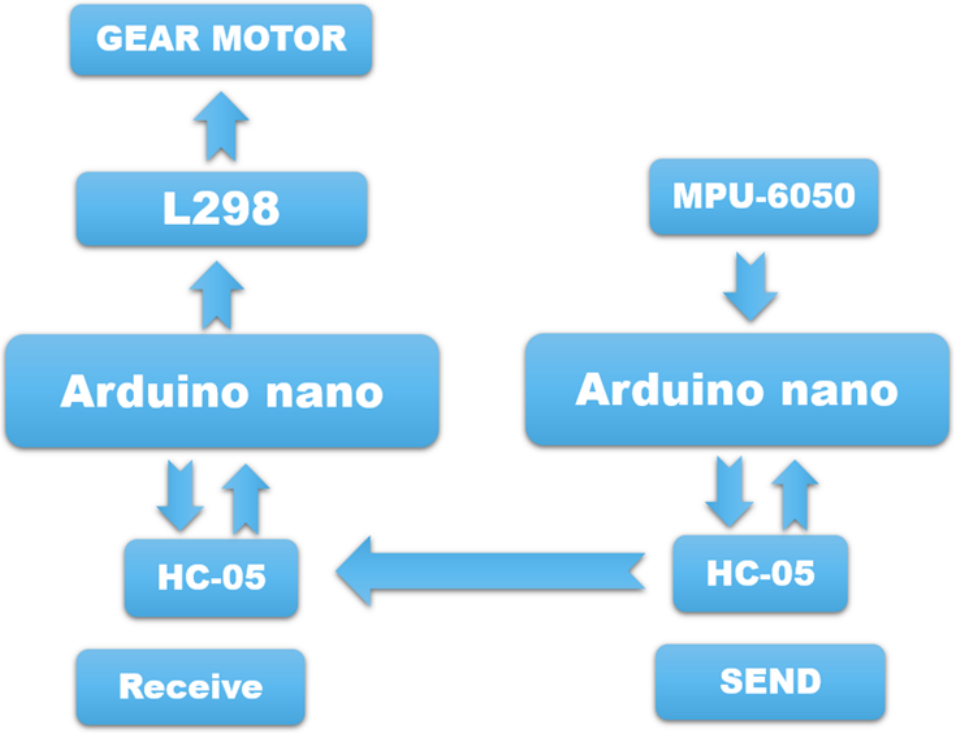


Figure 3.1: System Block Diagram of the Hand-Gesture-Controlled Differential Drive Robot

3.2 System Components and Selection

The system components were selected to match the functional requirements of a low-cost, wearable, and gesture-controlled differential drive robot. The project is divided into two main units: the transmitter unit, mounted on the glove, and the receiver unit, mounted on the robot car kit. Each component was chosen based on its function, availability, simplicity of integration, and suitability for an academic prototype. The Arduino Nano was used as the main controller in both units because it is compact and easy to program. The MPU-6050 sensor was selected to detect hand tilt and motion, while the HC-05 Bluetooth modules were used to provide wireless communication between the glove and the robot. The L298 motor

driver was used to control the two DC gear motors responsible for robot movement.

Table 3.1: Main Components of the Implemented Gesture-Controlled Robot

No.	Component	Quantity	Unit	Function	Reason for Selection
1	Arduino Nano	2	Transmitter and Receiver	Processes sensor data and controls system operation	Compact, low-cost, and suitable for embedded control
2	MPU-6050 Sensor	1	Transmitter	Detects hand tilt and motion	Combines accelerometer and gyroscope in one small module
3	HC-05 Bluetooth Module	2	Transmitter and Receiver	Provides wireless communication between glove and robot	Simple, available, and suitable for short-range communication
4	L298 Motor Driver	1	Receiver	Controls the direction and operation of the DC motors	Suitable for driving two DC motors
5	DC Gear Motors	2	Receiver	Drive the robot wheels	Provide the required motion for the car-kit robot
6	Robot Car Kit / Chassis	1	Receiver	Supports the robot structure and components	Ready-made, practical, and suitable for prototyping
7	Driving Wheels	2	Receiver	Convert motor rotation into robot motion	Required for differential drive movement

8	Free Caster Wheel	1	Receiver	Supports balance and smooth turning	Allows stable movement with two driving wheels
9	3.7V Lithium Battery	1	Transmitter	Powers the glove unit	Lightweight and suitable for wearable use
10	5V Power Bank / Boost Module	1	Transmitter	Provides stable voltage for the transmitter circuit	Converts battery voltage to a suitable level
11	9V Battery	1	Receiver	Powers the robot receiver unit and motors	Simple power source for prototype testing
12	Jumper Wires	Several	Both Units	Connect electronic components	Easy to use for circuit assembly
13	Glove	1	Transmitter	Holds the transmitter components on the user's hand	Enables wearable gesture control

3.3 Control Strategy

The control strategy of the system is based on converting hand tilt angles into motion commands for the differential drive robot. In the implemented prototype, the MPU-6050 sensor reads the hand movement, and the Arduino Nano calculates the approximate tilt angles around the X-axis and Y-axis. These angles are compared with a predefined threshold value. If the tilt angle exceeds the threshold, the transmitter sends a corresponding command through Bluetooth to the receiver unit.

The current implementation uses a threshold-based control strategy. This means that the robot does not continuously vary its speed based on every small angle change; instead, it sends discrete commands: **F** for forward, **B** for backward, **R** for right, **L**

for left, and **S** for stop. According to the code, forward and backward movements are driven using PWM values of approximately 125, while right and left turning commands use PWM values of approximately 185. This provides a simple and reliable control method suitable for the prototype stage.

Table 3.2: Gesture-Based Control Strategy

Hand Gesture / Condition	Command Sent	Motor Action	Robot Motion
(\theta_y > 45^\circ)	F	Both motors rotate forward	Move forward
(\theta_y < -45^\circ)	B	Both motors rotate backward	Move backward
(\theta_x > 45^\circ)	R	Right-turn motor logic activated	Turn right
(\theta_x < -45^\circ)	L	Left-turn motor logic activated	Turn left
Neutral hand position	S	Both motors stop	Stop

In the current software logic, the commands are checked sequentially. Therefore, if the hand produces both a forward tilt and a turning tilt at the same time, the system gives priority to the first satisfied condition in the code. For example, if the forward condition is detected first, the robot executes the forward command rather than combining forward and turning motion.

For future improvement, the control strategy can be upgraded from discrete command control to proportional control. In this improved approach, a small hand tilt would generate a low motor speed, while a larger tilt would generate a higher motor speed. The forward/backward tilt angle θy can be linked to the robot's linear velocity v, and the left/right tilt angle θx can be linked to the robot's angular velocity

ω. This would allow the robot to perform combined movements, such as moving forward while turning, instead of executing only one command at a time.

$$V = K_y \theta_y$$

$$\omega = K_x \theta_x$$

This strategy shows that the MPU-6050 readings can be directly related to robot speed and direction. However, for the implemented prototype, the simpler threshold-based control was selected because it is easier to implement, more stable during testing, and sufficient for demonstrating the basic gesture-controlled motion of the robot.

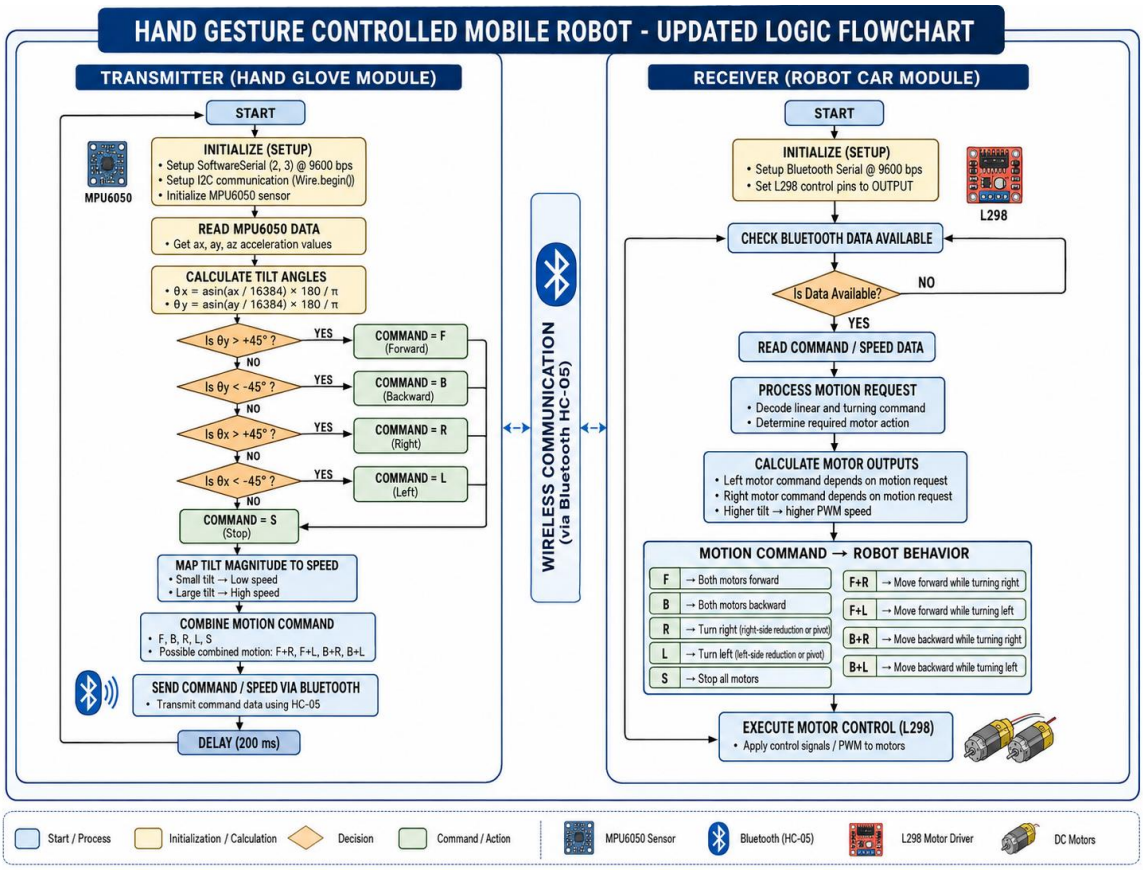


Figure 3.2: Logic Flowchart of the Gesture-Based Control Strategy

3.4 Electrical and Electronic Subsystems

The electrical and electronic subsystem integrates the sensors, microcontrollers, motor drivers, and power supply to enable gesture-based control of the differential drive robot.

3.4.1 Transmitter Unit (Hand Module):

- Components: Arduino Nano, MPU-6050 accelerometer/gyroscope sensor, HC-05 Bluetooth module, and 3.7V Li-ion battery.
- Function: Captures palm movements, processes sensor data, and transmits motion commands wirelessly.

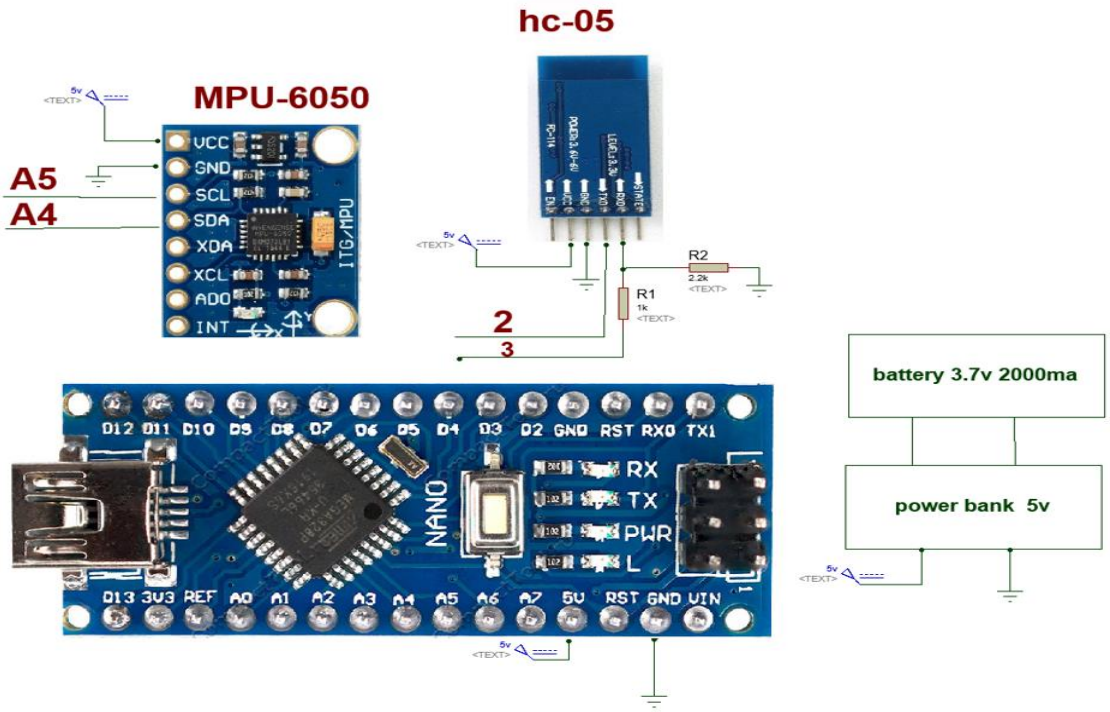


Figure 3.2: Wiring Diagram of the Transmitter Unit

3.4.2 Receiver Unit (Robot Module):

- Components: Arduino Nano, HC-05 Bluetooth module, L298 motor driver, two DC motors, and 9V battery.
- Function: Receives the wireless commands, interprets them, and actuates the motors for forward, backward, left, right, and stop movements.

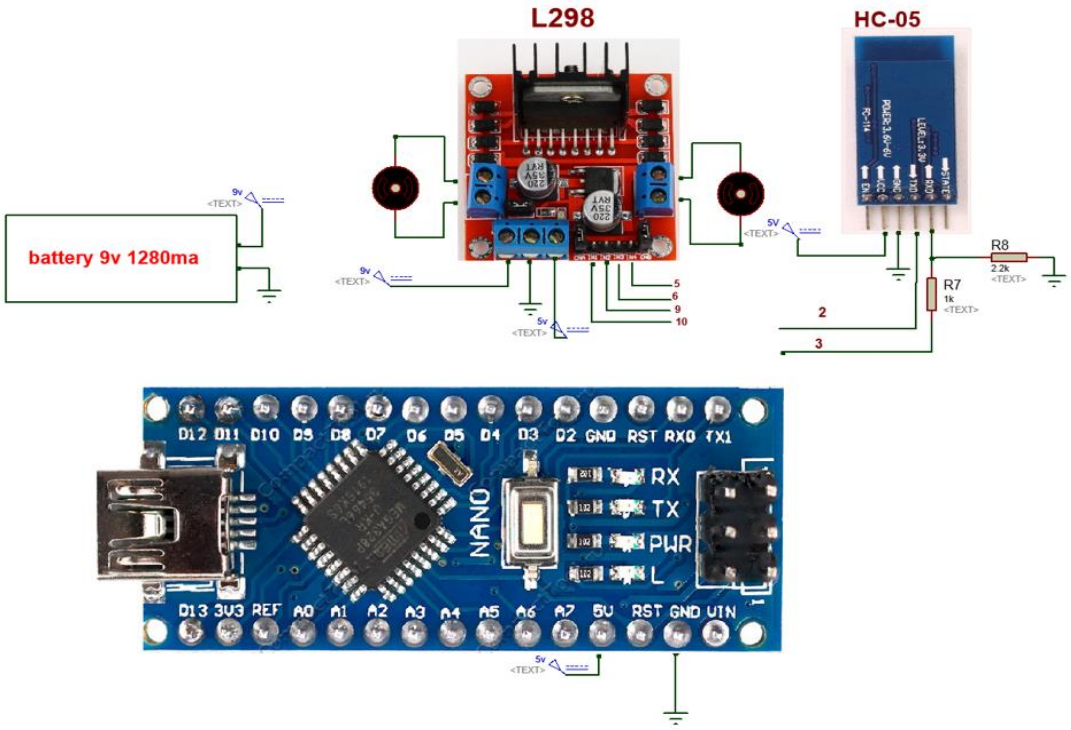


Figure 3.3: Wiring Diagram of the Receiver Unit

3.4.3 Communication and Protocols:

- I2C is used to read data from the MPU-6050 sensor.
- HC-05 Bluetooth modules provide low-latency wireless communication between the transmitter and receiver units.

3.5 Operating Scenarios

This section explains the main operating scenarios of the hand-gesture-controlled differential drive robot. These scenarios describe how the system behaves when the user performs different hand movements using the glove-based transmitter unit. Each scenario starts with reading the hand tilt through the MPU-6050 sensor, calculating the tilt direction, sending the corresponding command through the HC-05 Bluetooth module, and executing the required motion through the receiver unit and L298 motor driver.

Table 3.3: Operating Scenarios of the Gesture-Controlled Robot

Scenario	Hand Gesture	Sent Command	Robot Response
1	Hand tilted forward/upward	F	The robot moves forward by rotating both motors forward
2	Hand tilted backward/downward	B	The robot moves backward by rotating both motors backward
3	Hand tilted right	R	The robot turns right by changing the motor action on both sides
4	Hand tilted left	L	The robot turns left by changing the motor action on both sides
5	Hand in neutral position	S	The robot stops by turning off both motors
6	Small hand tilt	Low-speed motion	The robot can respond with lower speed in the improved proportional model
7	Large hand tilt	Higher-speed motion	The robot can respond with higher speed in the improved proportional model
8	Forward tilt + side tilt	Combined motion	The robot can move forward while turning in the improved control model

In the implemented prototype, the robot executes the basic commands separately: forward, backward, right, left, and stop. This means that the current software selects one main command according to the detected hand angle. However, the differential drive model also allows combined motion, such as moving forward while turning, by controlling the linear velocity and angular velocity together. This improvement can be applied in future versions by linking the magnitude of the MPU-6050 tilt angle to the motor PWM value, where a small tilt generates a lower speed and a larger tilt generates a higher speed.

3.6 Mathematical Modeling, Analysis, and Design

The mathematical modeling of the robot is based on the differential drive model, because the implemented robot uses two driving wheels and one free caster wheel. This model was selected because it matches the physical structure of the car-kit robot, requires only two motors, and allows the robot to move forward, backward, and rotate by controlling the speed difference between the right and left wheels. The free caster wheel does not generate motion; it only supports the robot body and improves stability during movement.

For a differential drive robot, the linear velocity of the robot vvv and the angular velocity ω\omegaω are determined by the velocities of the right and left wheels:

$$(1)\ V = (VR + VL) / 2$$

$$(2)\ \omega = (VR - VL) / L$$

Where:

V: Linear velocity of the robot

ω: Angular velocity of the robot

VR: Velocity of the right wheel

VL: Velocity of the left wheel

L: Distance between the two driving wheels

The robot position and heading can be represented by the following kinematic equations:

$$\dot{x} = v\cos(\theta) \quad (3)$$

$$\dot{Y} = v\sin(\theta) \quad (4)$$

$$\dot{\theta} = \omega \quad (5)$$

Where:

X: Robot position along the X-axis

Y: Robot position along the Y-axis

θ: Robot heading angle

dot{x}: Rate of change of position along X-axis

dot{Y}: Rate of change of position along Y-axis

dot{ θ }: Rate of change of robot heading angle

These equations explain the basic motion behavior of the robot. When both wheels rotate in the same direction and at the same speed, the robot moves forward or backward. When the wheel speeds are different, the robot turns. If one wheel is faster than the other, the robot follows a curved path. If one wheel stops while the other rotates, the robot turns around the slower side. Therefore, the differential drive model is suitable for this project because it directly represents the motion of the implemented two-wheel robot.

The hand gesture input is obtained from the MPU-6050 sensor. The sensor provides acceleration readings along the X, Y, and Z axes. In the implemented system, the X-axis and Y-axis readings are used to estimate the hand tilt angles. The tilt angles are calculated using the following equations:

(6) $$\theta_x = \sin^{-1}\left(\frac{a_x}{16384}\right) \times \frac{180}{\pi}$$

(7) $$\theta_y = \sin^{-1}\left(\frac{a_y}{16384}\right) \times \frac{180}{\pi}$$

Where:

θx: Estimated hand tilt angle around the X-axis

θy: Estimated hand tilt angle around the Y-axis

ax: Raw acceleration reading on the X-axis

ay: Raw acceleration reading on the Y-axis

16384: MPU-6050 accelerometer scale factor at ±2g

π: Mathematical constant pi

The value 16384 is used because the MPU-6050 accelerometer gives approximately 16384 LSB/g when it operates in the ±2g range [3]. Therefore, a raw acceleration value can be converted into a normalized acceleration value before estimating the tilt angle.

For example, if the MPU-6050 reading is approximately:

ay = 12000

Then:

$$\frac{12000}{16384} = 0.732$$

$$\theta_y = \sin^{-1}(0.732) \times \frac{180}{\pi} \approx 47^\circ$$

This means that a raw MPU-6050 value of about 12000 corresponds approximately to a hand tilt angle of 47°, which is close to the threshold used for motion detection. This explains why a threshold around 45° is suitable for distinguishing intentional hand gestures from small hand movements.

The control method used in the implemented prototype is a threshold-based control algorithm**.** The Arduino compares the calculated hand angles with predefined angle limits. If the angle exceeds the threshold, the corresponding motion command is sent to the robot.

$$\theta y > 45\circ \Rightarrow F \Rightarrow vR = vL > 0$$

$$\theta y < -45\circ \Rightarrow B \Rightarrow vR = vL < 0$$

$$\theta x > 45\circ \Rightarrow R \Rightarrow vR > vL$$

$$\theta x < -45\circ \Rightarrow L \Rightarrow vL > vR$$

$$|\theta x| \leq 45\circ \text{ and } |\theta y| \leq 45\circ \Rightarrow S \Rightarrow vR = vL = 0$$

In this model, the command **F** means that both motors receive forward driving signals, so the robot moves forward. The command **B** reverses both motors, so the robot moves backward. The commands **R** and **L** generate turning motion by creating a speed difference between the two wheels. The command **S** stops both motors.

If the hand gives both a forward and turning gesture at the same time, the differential drive model can represent this as a combined linear and angular motion:

$$(8)\ V = K_y \theta_y$$

$$(9)\ \omega = K_x \theta_x$$

Where:

K_y: Proportional constant relating forward/backward hand tilt to robot linear speed

K_x: Proportional constant relating left/right hand tilt to robot angular speed

θ_y: Forward/backward hand tilt angle

θ_x: Left/right hand tilt angle

This means that a small tilt angle produces a lower speed, while a larger tilt angle produces a higher speed. In the current prototype, the implemented software uses a simplified version of this idea by sending discrete commands **F, B, R, L, and S** based on the angle threshold. However, the mathematical model shows how the system can be improved to support proportional speed control and combined forward-turning motion in future development.

From the electrical and electronic design perspective, the transmitter unit consists of an Arduino Nano, MPU-6050 sensor, HC-05 Bluetooth module, and a 3.7V lithium battery with a 5V power module. The receiver unit consists of another Arduino Nano, an HC-05 Bluetooth module, an L298 motor driver, two DC gear motors, and a 9V battery. The Arduino Nano was selected because it is compact, low-cost, and suitable for embedded control. The MPU-6050 was selected because it can detect hand orientation and tilt, making it suitable for a wearable gesture-control system. The HC-05 modules were selected for simple wireless serial

communication, while the L298 motor driver was used to control the direction of the DC motors.

Overall, the selected design is suitable for the project requirements because it provides a simple, low-cost, and practical implementation of hand-gesture-based robot control. The differential drive model justifies the robot movement, the MPU-6050 equations justify the gesture detection process, and the threshold-based control logic explains how hand tilt is converted into motor commands.

3.7 Chapter Conclusion

This chapter presented the analysis and design of the hand-gesture-controlled differential drive robot. It explained the overall system architecture, where the project is divided into a glove-based transmitter unit and a robot-mounted receiver unit. The transmitter reads the user's hand motion using the MPU-6050 sensor and sends commands through HC-05 Bluetooth, while the receiver processes these commands and controls the motors through the L298 motor driver.

The chapter also introduced the mathematical model of the differential drive robot and explained why this model is suitable for a robot with two driving wheels and one free caster wheel. The motion equations showed how the linear and angular velocities of the robot depend on the velocities of the right and left wheels. In addition, the MPU-6050 angle equations were used to explain how hand tilt is converted into motion commands.

The main system components were identified and justified, including the Arduino Nano, MPU-6050, HC-05 Bluetooth modules, L298 motor driver, DC gear motors, batteries, and robot car kit. The control strategy was also described as a threshold-based method, where specific hand angles generate motion commands such as forward, backward, right, left, and stop. Finally, the hardware connection design

was presented through the transmitter and receiver circuits. Overall, this chapter established the technical foundation required for the implementation and testing stage discussed in the next chapter.

Chapter 4

Implementation and Testing

4.1 System Implementation

The implemented system consists of two main parts: the transmitter unit and the receiver unit**.** The transmitter unit was built as a glove worn on the user's hand. It includes an Arduino Nano, an MPU-6050 sensor, an HC-05 Bluetooth module, and a small lithium battery with a power module. During operation, the MPU-6050 reads the hand tilt values, and the Arduino Nano processes these readings to determine the required movement command. The command is then sent wirelessly to the robot through the HC-05 Bluetooth module.

The receiver unit was mounted on a robot car kit with two driving wheels and one free caster wheel. It consists of an Arduino Nano, an HC-05 Bluetooth module, an L298 motor driver, two DC gear motors, and a 9V battery. The receiver Arduino reads the command received through Bluetooth and sends the suitable control signals to the L298 motor driver. According to the received command, the motor driver operates the motors to move the robot forward, backward, right, left, or stop.

The implementation followed the design presented in Chapter 3. The transmitter unit is responsible for gesture detection and command transmission, while the receiver unit is responsible for command interpretation and motor control.

4.2 Practical Testing Results

The practical testing was performed to evaluate the robot's response to the main gesture commands and to verify the stability of wireless communication between the transmitter glove and the receiver robot. The robot was tested on a flat indoor surface at an approximate distance of **1 meter** between the user and the robot. The tested commands included **Forward, Backward, Right, Left, and Stop**. Since the current implementation uses a threshold-based control strategy, each gesture

produces one dominant command at a time, and combined motion such as forward while turning was not implemented in the current prototype.

A total of **50 testing trials** were considered, with **10 trials for each command**. A trial was considered successful when the robot responded to the correct hand gesture and executed the expected movement without command failure. The response time was generally observed between **1 and 2 seconds**, which is acceptable for the prototype stage. The robot does not include obstacle detection; therefore, if an obstacle is placed in front of it, the robot does not stop automatically and may collide with the obstacle. This limitation is considered in the system evaluation and future work.

Table 4.1: Practical Testing Results of Gesture-Based Commands

Command	Hand Gesture Condition	Number of Trials	Successful Trials	Success Rate	Observed Robot Behavior
Forward	(\theta_y > 45^\circ)	10	9	90%	Robot moved forward correctly
Backward	(\theta_y < -45^\circ)	10	9	90%	Robot moved backward correctly
Right	(\theta_x > 45^\circ)	10	8	80%	Robot turned right correctly
Left	(\theta_x < -45^\circ)	10	8	80%	Robot turned left correctly
Stop	Neutral hand position	10	10	100%	Robot stopped correctly
Total	—	**50**	**44**	**88%**	Overall system response was acceptable

The overall success rate was calculated using the following equation:

(10) Success Rate = Total Number of Trials / Number of Successful Trials * 100

$$\text{Success Rate} = 50 / 44 \times 100 = 88\%$$

The results show that the robot was able to execute all basic movement commands with an overall success rate of **88%**. The stop command achieved the highest reliability because it depends on returning the hand to the neutral position. Turning commands showed slightly lower success rates compared with forward and backward motion, mainly because side tilting of the hand requires better sensor calibration and more stable hand positioning. Overall, the prototype achieved acceptable performance for an academic implementation, while still leaving room for improvement in sensor calibration, response time, and motion smoothness.

4.3 Comparative Analysis with Existing Systems

A comparative analysis was conducted to evaluate the implemented prototype against the gesture-controlled robot systems discussed in the literature review. Since exact commercial costs and measured speed values were not available for all systems, the comparison is presented using realistic qualitative levels such as **low**, **medium**, and **high**. This approach is more appropriate than using unsupported numerical values. The comparison focuses on the main criteria requested for system evaluation: communication method, speed behavior, accuracy, cost level, complexity, and limitations.

Table 4.2: Comparative Analysis with Existing Gesture-Controlled Robot Systems

System / Study	Main Technology	Communication	Speed Behavior	Accuracy / Reliability	Cost Level	Main Limitations
Vala, 2018 [1]	Arduino Uno + ADXL335 accelerometer	RF module	Basic fixed motion commands	Reliable for basic gestures	Low	Uses older accelerometer technology and RF communication
Rohan et al., 2024 [2]	MPU-6050 / camera-based and IoT-based methods	MQTT / IoT	Can support real-time control	Higher accuracy but depends on system complexity	Medium to High	More complex, may require higher processing power
Implemented Project	Arduino Nano + MPU-6050 + HC-05 + L298	Bluetooth HC-05	Basic prototype motion; speed not precisely measured	88% based on 50 practical trials	Low	Limited Bluetooth range, no obstacle avoidance, no combined motion in current code

The implemented project provides a low-cost and simple solution compared with more advanced IoT or vision-based systems. Its main advantage is that it uses available components and a direct control method that is easy to build, test, and understand. However, its performance is still limited compared with more advanced systems because the current prototype uses discrete commands rather

than proportional speed control. It also does not include obstacle detection, meaning that the robot will not stop automatically if an obstacle is placed in front of it. Despite these limitations, the system achieved acceptable performance for an academic prototype and successfully demonstrated the main concept of controlling a differential drive robot using hand gestures.

4.4 Performance Analysis and System Evaluation

The performance of the implemented system was evaluated based on the practical testing results and the main technical requirements defined for the project. The evaluation focused on the robot's ability to respond correctly to hand gestures, maintain stable wireless communication, execute the required movement commands, and meet the expected prototype-level performance. The tested commands included forward, backward, right, left, and stop, with a total of 50 practical trials.

The system achieved an overall success rate of **88%**, calculated from 44 successful trials out of 50. This result shows that the robot was able to perform the required motion commands with acceptable reliability for an academic prototype. The response time was observed to be approximately **1–2 seconds**, which satisfies the basic requirement for gesture-based control at the prototype level. The Bluetooth communication was stable within the tested distance of approximately **1 meter**, and the robot movement was generally smooth during normal operation.

However, the evaluation also showed some limitations. The robot does not include an obstacle detection system, so it does not stop automatically when an object is placed in its path. In addition, the current implementation does not support combined motion, such as moving forward while turning, because the software executes one dominant command at a time. The robot speed was not measured using precise instruments, so it was evaluated qualitatively as a low-speed prototype

suitable for indoor testing. Overall, the implemented system meets the main project objectives and most of the technical requirements, while leaving clear opportunities for future improvement in speed control, obstacle avoidance, Bluetooth range, and gesture accuracy.

Table 4.3: Evaluation of the Implemented System Against Technical Requirements

Evaluation Criterion	Required / Expected Performance	Achieved Performance	Evaluation
Gesture Recognition	Detect basic hand gestures	Forward, backward, right, left, and stop detected	Achieved
Motion Commands	Execute five main commands	All five commands were executed	Achieved
Response Time	Acceptable prototype response	Approximately 1–2 seconds	Achieved
Communication	Stable short-range wireless link	Stable at approximately 1 meter	Achieved
Accuracy / Reliability	Reliable command execution	88% success rate based on 50 trials	Achieved
Motion Stability	Stable movement on flat surface	Movement was generally stable	Achieved
Obstacle Handling	Detect or avoid obstacles	Not implemented; robot may collide with obstacles	Not Achieved
Combined Motion	Forward/backward with turning	Not implemented in current code	Not Achieved
Speed Measurement	Defined measured speed	Not precisely measured; low-speed prototype behavior	Partially Achieved

This evaluation confirms that the implemented robot satisfies the essential functional goals of the project. It successfully demonstrates gesture-based control, wireless command transmission, and differential-drive motion. At the same time,

the limitations identified in this section provide a clear direction for future development.

4.5 Strengths and Limitations

A self-evaluation of the implemented system shows that the project has several clear strengths as well as some practical limitations. One of the main strengths is the simplicity of the control method. The robot can be controlled using natural hand gestures without the need for a joystick or wired controller. This makes the system easy to understand and suitable as an introductory prototype for gesture-based human–robot interaction. Another strength is the use of low-cost and commonly available components such as Arduino Nano, MPU-6050, HC-05 Bluetooth, and L298 motor driver. This makes the system affordable, easy to reproduce, and suitable for academic implementation.

The system also successfully achieved the basic required functions. It was able to read hand gestures, transmit commands wirelessly, and execute the main robot movements: forward, backward, right, left, and stop. The testing results showed an overall success rate of **88%**, which is acceptable for a prototype system. The Bluetooth connection was stable during the short-range indoor tests, and the robot movement was generally smooth on a flat surface.

However, the system still has several limitations that must be clearly acknowledged. The robot does not include obstacle detection, so it cannot stop automatically when an object is in front of it. If an obstacle exists in its path, the robot may collide with it. The system also does not support combined motion in the current implementation, meaning that it cannot move forward and turn at the same time using the existing code logic. In addition, the robot speed was not measured using precise instruments, so the speed evaluation remains qualitative rather than numerical. The Bluetooth range is also limited, and the gesture recognition depends

on the stability of the user's hand position and the calibration of the MPU-6050 sensor.

Overall, the implemented system is successful as an academic prototype, but it is not yet a complete practical assistive robot. Its strengths prove that the main concept works, while its limitations show the areas that need improvement in future development, especially obstacle avoidance, proportional speed control, longer communication range, and better sensor calibration.

4.6 Key Performance Indicators (KPIs)

The performance of the implemented system was evaluated using a set of practical Key Performance Indicators (KPIs). These indicators were selected based on the project objectives and the main challenges discussed in previous gesture-controlled robot studies, such as response time, accuracy, reliability, and ease of use [1][2]. Since the project is an academic prototype, the KPIs focus on measurable practical behavior rather than commercial-level performance.

Table 4.4: Key Performance Indicators of the Implemented System

KPI	Evaluation Method	Target / Expected Value	Achieved Result	Status
Command Success Rate	50 practical trials	≥ 85%	88%	Achieved
Response Time	Observation during testing	≤ 2 seconds	1–2 seconds	Achieved
Bluetooth Range	Indoor test distance	Around 1 meter minimum	≈ 1 meter	Achieved
Gesture Recognition	Testing F, B, R, L, S commands	Detect all basic commands	All commands detected	Achieved
Motion Stability	Observation on flat surface	Stable movement	Generally stable	Achieved

User Accessibility	Simplicity of control method	Easy hand-gesture control	Simple and intuitive	Achieved
Obstacle Handling	Testing system behavior near obstacles	Automatic stop preferred	Not implemented	Not Achieved
Combined Motion	Forward + turning command	Supported in advanced model	Not implemented in current code	Not Achieved

The results show that the system achieved the main KPIs required for a functional gesture-controlled robot prototype. The overall command success rate reached **88%**, which indicates acceptable reliability for basic movement commands. The response time was approximately **1–2 seconds**, which satisfies the prototype requirement and supports real-time interaction at a basic level. The robot also demonstrated stable movement on a flat surface and successfully responded to the five main commands: forward, backward, right, left, and stop.

However, some KPIs were not fully achieved. The current prototype does not include obstacle detection, so it cannot stop automatically when an obstacle is detected. Also, combined motion, such as moving forward while turning, is not implemented in the current code because the system executes one dominant command at a time. These results are realistic and consistent with the limitations discussed in previous studies, where latency, accuracy, hardware constraints, and environmental limitations are considered major challenges in gesture-controlled robotic systems [2]. Overall, the KPI evaluation confirms that the system meets the main academic objectives while clearly identifying areas for future improvement.

Chapter 5

Conclusion and Recommendations

5.1 Project Summary and Main Achievements

This project presented the design and implementation of a hand-gesture-controlled differential drive robot. The system was developed using a glove-based transmitter unit and a robot-mounted receiver unit. The transmitter reads the user's hand movement using the MPU-6050 sensor, converts the detected tilt into motion commands, and sends these commands wirelessly through the HC-05 Bluetooth module. The receiver unit processes the received commands and controls the robot motors through the L298 motor driver.

The main achievement of the project is the successful implementation of a simple and intuitive gesture-based control system. The robot was able to perform the main motion commands: forward, backward, right, left, and stop. In addition, the project successfully integrated sensors, wireless communication, embedded programming, and motor control into one working prototype. The testing results showed that the system achieved acceptable performance for an academic prototype, with an overall command success rate of 88% and a response time of approximately 1–2 seconds.

5.2 Difficulties and Challenges

During the development of the project, several technical, organizational, and time-related challenges were faced. From a technical perspective, one of the main challenges was obtaining stable and meaningful readings from the MPU-6050 sensor. Since the system depends on hand tilt angles, small changes in hand position could affect the command sent to the robot. Therefore, the gesture threshold had to be selected carefully to reduce false commands and improve the reliability of movement detection.

Another technical challenge was ensuring stable Bluetooth communication between the transmitter glove and the receiver robot. The system depends on wireless transmission, so any delay or weak connection could affect the response of the robot. In addition, the wiring of the Arduino Nano, HC-05 Bluetooth modules, L298 motor driver, and DC motors required careful checking to avoid wrong connections or unstable motor behavior.

From an organizational perspective, the project required coordination between the transmitter unit and receiver unit, since each part had different hardware and software responsibilities. The transmitter was responsible for reading the hand gesture and sending the command, while the receiver was responsible for interpreting the command and controlling the motors. Time was also a limitation because the system had to be assembled, programmed, tested, and improved within the available project period.

Overall, these challenges were handled through step-by-step implementation and repeated testing. Each subsystem was checked separately before testing the full system. This helped identify errors more easily and improved the final stability of the prototype.

5.3 Future Work

If the project continues for another academic year, several improvements can be added to make the system more practical, safer, and closer to a real assistive robotic platform. The first improvement is adding an obstacle detection system using ultrasonic or infrared sensors. This would allow the robot to detect objects in front of it and stop automatically instead of colliding with obstacles. This feature is important because the current prototype depends completely on the user's control and does not react to obstacles by itself.

Another future improvement is developing the control strategy from simple threshold-based commands into proportional speed control. In the current prototype, the robot receives fixed commands such as forward, backward, right, left, and stop. In a more advanced version, the tilt angle from the MPU-6050 can be directly linked to motor speed, so a small hand tilt produces slow movement and a larger tilt produces faster movement. This would make the robot smoother, more responsive, and easier to control.

The system can also be improved by supporting combined motion, such as moving forward while turning left or right. This requires modifying the software so that the robot can process both linear and angular commands at the same time. In addition, the Bluetooth communication range can be improved by using a stronger wireless module or replacing Bluetooth with Wi-Fi. The glove can also be redesigned to be lighter, more comfortable, and more stable on the user's hand.

Finally, a robotic arm can be added to expand the robot's assistive function beyond movement. With a robotic arm, the robot could perform simple tasks such as holding or carrying lightweight objects. These improvements would transform the project from a basic gesture-controlled mobile robot into a more complete assistive robotic system.

5.4 Recommendations

Based on the results and limitations of this project, several recommendations can be proposed for future students, developers, and institutions interested in adopting or improving this system. Future students should begin by testing each subsystem separately before integrating the full system. For example, the MPU-6050 sensor readings, Bluetooth communication, and motor driver circuit should each be verified individually. This approach reduces errors during final assembly and makes troubleshooting easier.

It is also recommended to calibrate the MPU-6050 sensor carefully before testing, because inaccurate angle readings may lead to wrong motion commands. Students working on similar projects should also document every test clearly, including the number of trials, success rate, response time, and observed problems. This makes the evaluation more reliable and prevents the results from appearing general or unsupported.

For future development, institutions or project teams should consider adding safety features such as obstacle detection, automatic stopping, and improved power management. If the system is intended for assistive applications, the glove should be redesigned to be lighter, more comfortable, and easier to wear. In addition, using a stronger wireless communication module and proportional speed control would improve the robot's responsiveness and usability. Overall, the project is recommended as a strong starting point for students interested in embedded systems, mobile robotics, and human–robot interaction, but it should be further developed before being used in real assistive environments.

Appendices

Appendix A: Source Codes

Appendix A.1: Transmitter Unit Code

This code is used in the transmitter unit mounted on the glove. It reads the hand tilt values from the MPU-6050 sensor, calculates the approximate hand angles, and sends movement commands through the HC-05 Bluetooth module. The sent commands are: **F** for forward, **B** for backward, **R** for right, **L** for left, and **S** for stop.

```
#define RX 2

#define TX 3



#include <Wire.h>

#include <MPU6050.h>

#include <SoftwareSerial.h>



SoftwareSerial mySerial(RX, TX); // RX, TX

MPU6050 mpu;



int16_t ax, ay, az;



void setup() {
```

```
  mySerial.begin(9600);
  Wire.begin();
  mpu.initialize();
}

void loop() {
  mpu.getAcceleration(&ax, &ay, &az);

  float angleX = asin(ax / 16384.0) * 180.0 / PI;
  float angleY = asin(ay / 16384.0) * 180.0 / PI;

  if      (angleY > 45)  mySerial.println("F");
  else if (angleY < -45) mySerial.println("B");
  else if (angleX > 45)  mySerial.println("R");
  else if (angleX < -45) mySerial.println("L");
  else                   mySerial.println("S");

  delay(200);
```

}

Appendix A.2: Receiver Unit Code

This code is used in the receiver unit mounted on the robot. It receives movement commands from the transmitter through the HC-05 Bluetooth module and controls the DC motors through the L298 motor driver. According to the received command, the robot moves forward, backward, turns right, turns left, or stops.

```
#define RX 2

#define TX 3



#define IN1 10

#define IN2 9

#define IN3 6

#define IN4 5



#include <SoftwareSerial.h>



SoftwareSerial mySerial(RX, TX);



void setup() {
```

```
  mySerial.begin(9600);


  pinMode(IN1, OUTPUT);
  pinMode(IN2, OUTPUT);
  pinMode(IN3, OUTPUT);
  pinMode(IN4, OUTPUT);
}


void loop() {
  if (mySerial.available()) {
    char data = mySerial.read();


    if (data == 'F') {
      analogWrite(IN1, 125);
      digitalWrite(IN2, 0);


      analogWrite(IN3, 125);
      digitalWrite(IN4, 0);
```

```
}


else if (data == 'B') {
  digitalWrite(IN1, 0);
  analogWrite(IN2, 125);

  digitalWrite(IN3, 0);
  analogWrite(IN4, 125);
}

else if (data == 'R') {
  analogWrite(IN1, 185);
  digitalWrite(IN2, 0);

  digitalWrite(IN3, 0);
  digitalWrite(IN4, 0);
}
```

```
    else if (data == 'L') {
      digitalWrite(IN1, 0);
      digitalWrite(IN2, 0);

      analogWrite(IN3, 185);
      digitalWrite(IN4, 0);
    }

    else if (data == 'S') {
      digitalWrite(IN1, 0);
      digitalWrite(IN2, 0);

      digitalWrite(IN3, 0);
      digitalWrite(IN4, 0);
    }
  }
}
```

Appendix B: User Manual

B.1 System Installation

To install the system, the transmitter unit should be fixed on the user's hand using the glove. The MPU-6050 sensor must be positioned properly so that it can detect the palm tilt clearly. The receiver unit should be mounted on the robot car chassis, with the Arduino Nano, HC-05 Bluetooth module, L298 motor driver, DC motors, and battery connected according to the wiring diagrams presented in Chapter 3.

B.2 System Operation

After connecting the batteries to both the transmitter and receiver units, the Bluetooth modules should establish communication between the glove and the robot. Once the system is powered on, the user can control the robot by moving the palm in different directions. The robot responds according to the detected gesture.

B.3 Control Commands

Table B.1: User Control Commands for the Gesture-Controlled Robot

Hand Gesture	Command	Robot Action
Hand tilted upward	F	Move Forward
Hand tilted downward	B	Move Backward
Hand tilted right	R	Turn Right
Hand tilted left	L	Turn Left
Neutral position	S	Stop

B.4 Safety Notes

The robot should be tested on a flat surface to avoid unstable movement. The batteries must be checked before operation to ensure stable power. The system should not be used near stairs, obstacles, or wet areas. If the robot does not respond correctly, the user should stop the system and check the Bluetooth connection, wiring, and battery level.

References

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[2] Rohan V., Sai Sudheer K., Shreedhar Chandrashekar Malagamani, and Dhineshbabu N. R., “Hand Gesture Control Robot,” *International Journal of Scientific Research and Engineering Development (IJSRED)*, vol. 7, no. 6, pp. 809–811, Nov.–Dec. 2024.

[3] InvenSense, “MPU-6000 and MPU-6050 Register Map and Descriptions,” Revision 3.2, Nov. 2011.